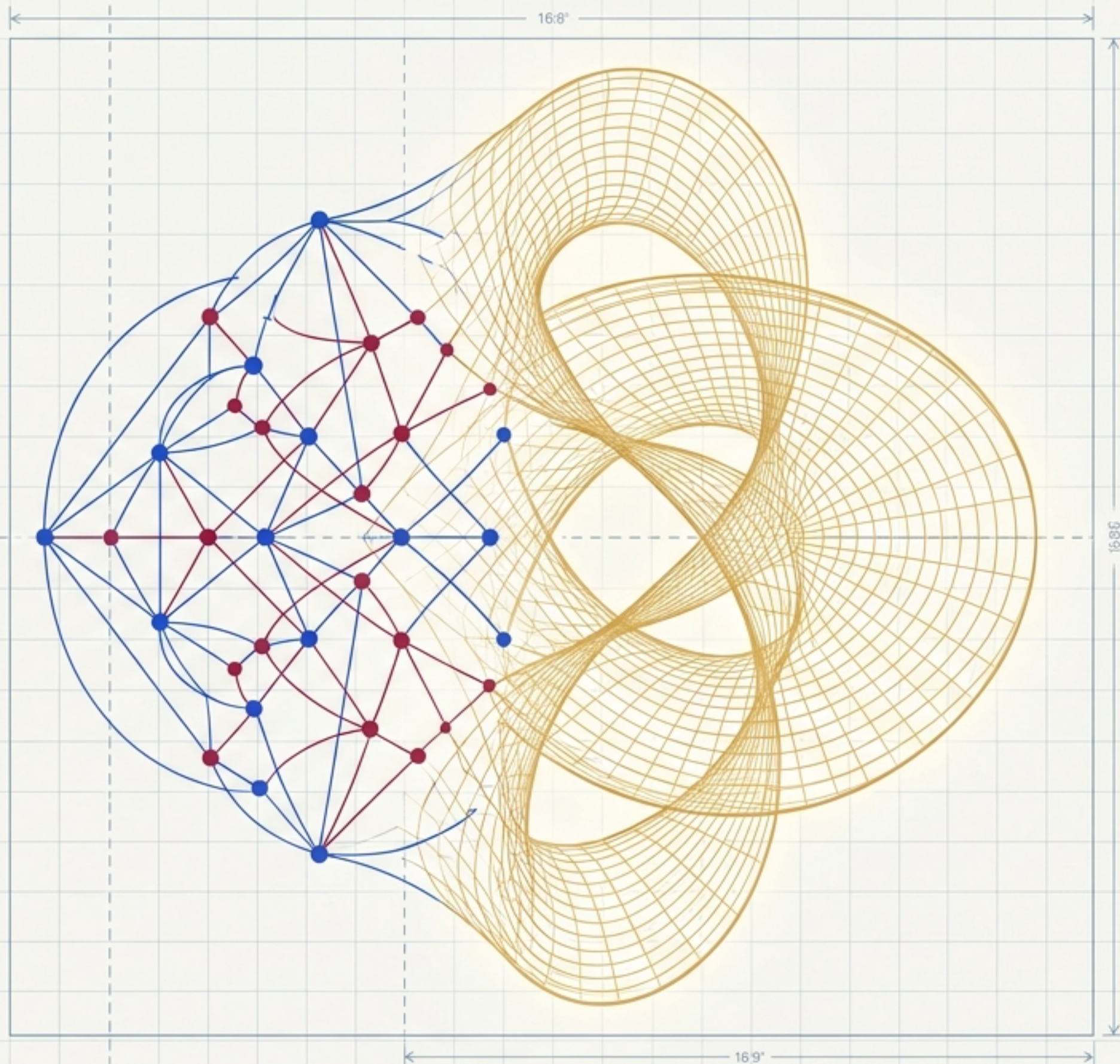


An 80-Year-Old Mathematical Wall, Breached.

Unpacking the September 2026
OpenAI Proof: A Torsion-Free
Group Algebra with Zero Divisors

- Group Elements
- Field Elements
- Topological Structures



The Fall of Kaplansky's Zero-Divisor Conjecture

Traced back to Graham Higman (1940) and Irving Kaplansky (1956), the conjecture survived decades of scrutiny.

Historical Context

In September 2026, an AI-generated, machine-checked paper entirely disproved it.

Modern Disproof



LEAN FORMALIZED /
MACHINE CHECKED

Theorem 1.1: *There exist a **finitely presented torsion-free group** G and nonzero elements $\alpha, \beta \in \mathbb{F}_2[G]$ such that $\alpha\beta = 0$.*

Presentation

Ring of
Coefficients

Zero Divisor
Condition

Group Properties

```
import TorsionFreeZeroDivisors.lean
```

Verification File

Defining the Arena: The Group Algebra $\mathbb{K}[G]$



Example Multiplication

$$(k_1g_1 + k_2g_2) \times (k_3g_3) = (k_1k_3)(g_1g_3) + (k_2k_3)(g_2g_3)$$



The Core Rule

A "zero divisor" occurs when two strictly non-zero elements multiply to perfectly cancel each other out, equaling exactly 0.

The Torsion Loophole: Why Elements of Finite Order Must Be Excluded

Let $g^m = 1$ (where $m > 1$)



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MACHINE CHECKED

$$(1 - g)(1 + g + g^2 + \dots + g^{m-1}) = 0$$

$$1 + g + g^2 + \dots + g^{m-1} - g - g^2 - \dots - g^{m-1} - g^m$$

$$1 - g^m$$

$$\downarrow g^m = 1$$

$$1 - 1 = 0$$

Takeaway: Finite-order elements (torsion) automatically create zero divisors. Kaplansky explicitly asked: does removing torsion remove all zero divisors?

The Three Suspects: Kaplansky's Conjectures for Torsion-Free Groups

	Conjecture	Defining Equation	Historical Status
 Logical Cascade: Unit implies Zero-Divisor implies Idempotent.	Unit Conjecture (Only trivial units kg)	$ab = 1$ (where a, b are not just trivial group elements)	 Disproved (Gardam, 2021)
	Zero-Divisor Conjecture (No nontrivial zero divisors)	$ab = 0$ (where $a \neq 0$ and $b \neq 0$)	 Disproved (OpenAI, 2026)
	Idempotent Conjecture (Only idempotents 0 and 1)	$e^2 = e$	 Still Open

Insight: Finding a nontrivial idempotent 'e' automatically gives the zero divisor $e(1-e)=0$.
Breaking the zero-divisor conjecture pushes mathematics right to the edge of the final mystery.

The First Crack: Gardam Disproves the Unit Conjecture (2021)

Giles Gardam found a counterexample using the Hantzsche-Wendt group ($\mathbb{P}$) and the field of two elements ($\mathbb{F}_2$).

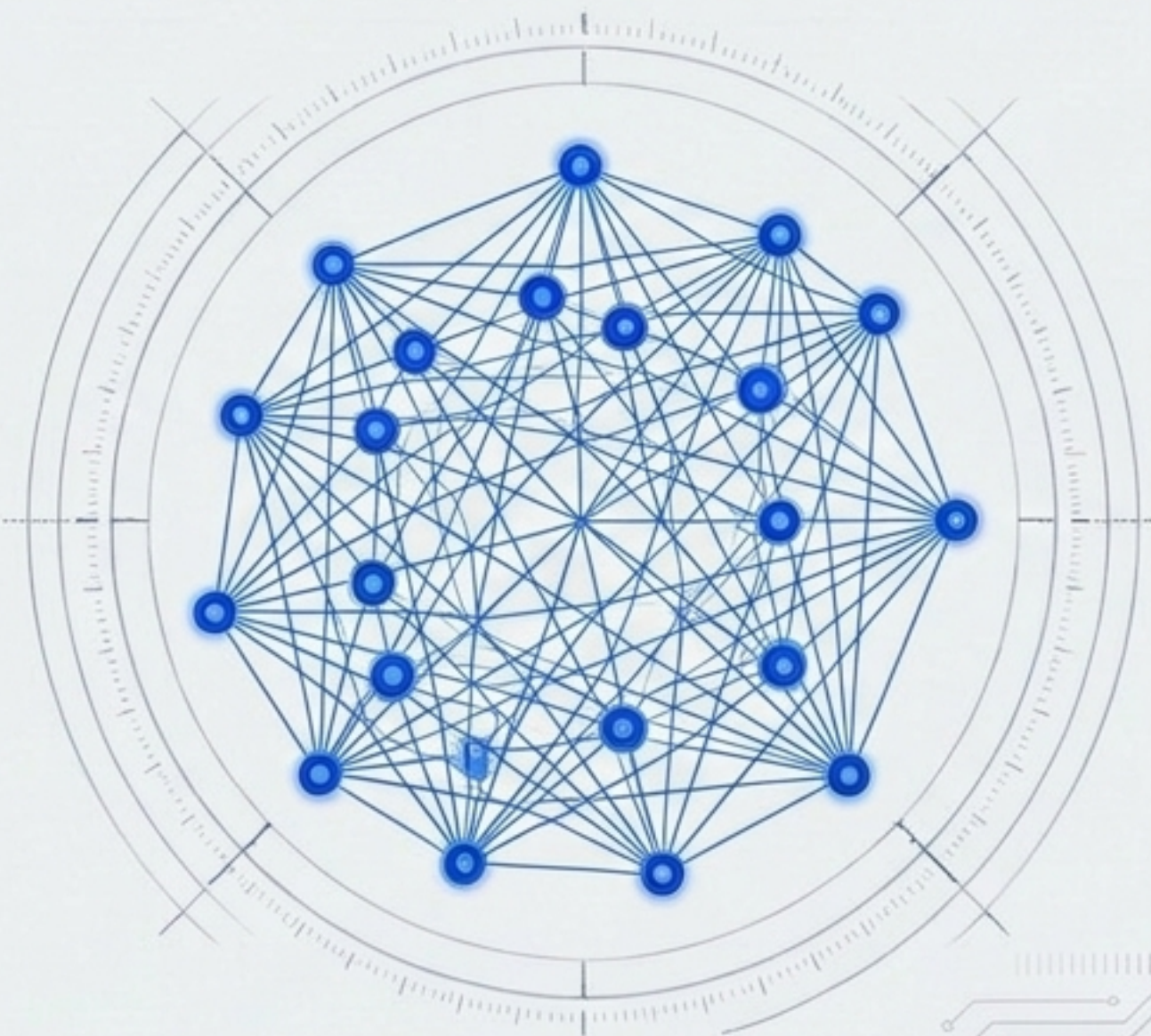
The $\mathbb{F}_2$ Rule: Characteristic 2 Parity

$$1 + 1 = 0$$

Over $\mathbb{F}_2$, a product is zero exactly when every group element occurs an even number of times.

Parity is everything.

Structural Map of the 21-Term Unit



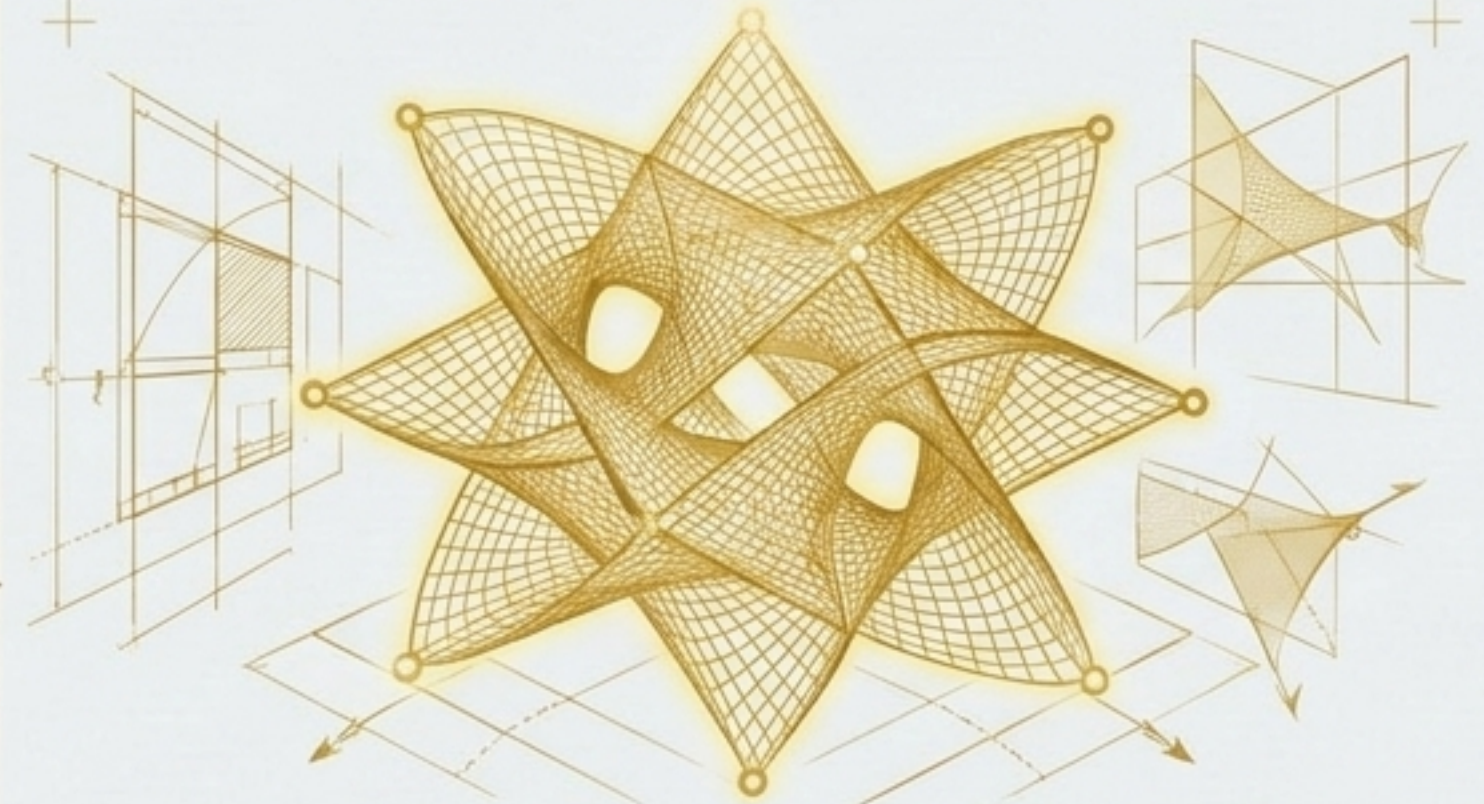
The 2026 Strategy: An Algebraic Goal via a Geometric Mechanism

The Algebraic Goal

$$\alpha \cdot \beta = 0$$

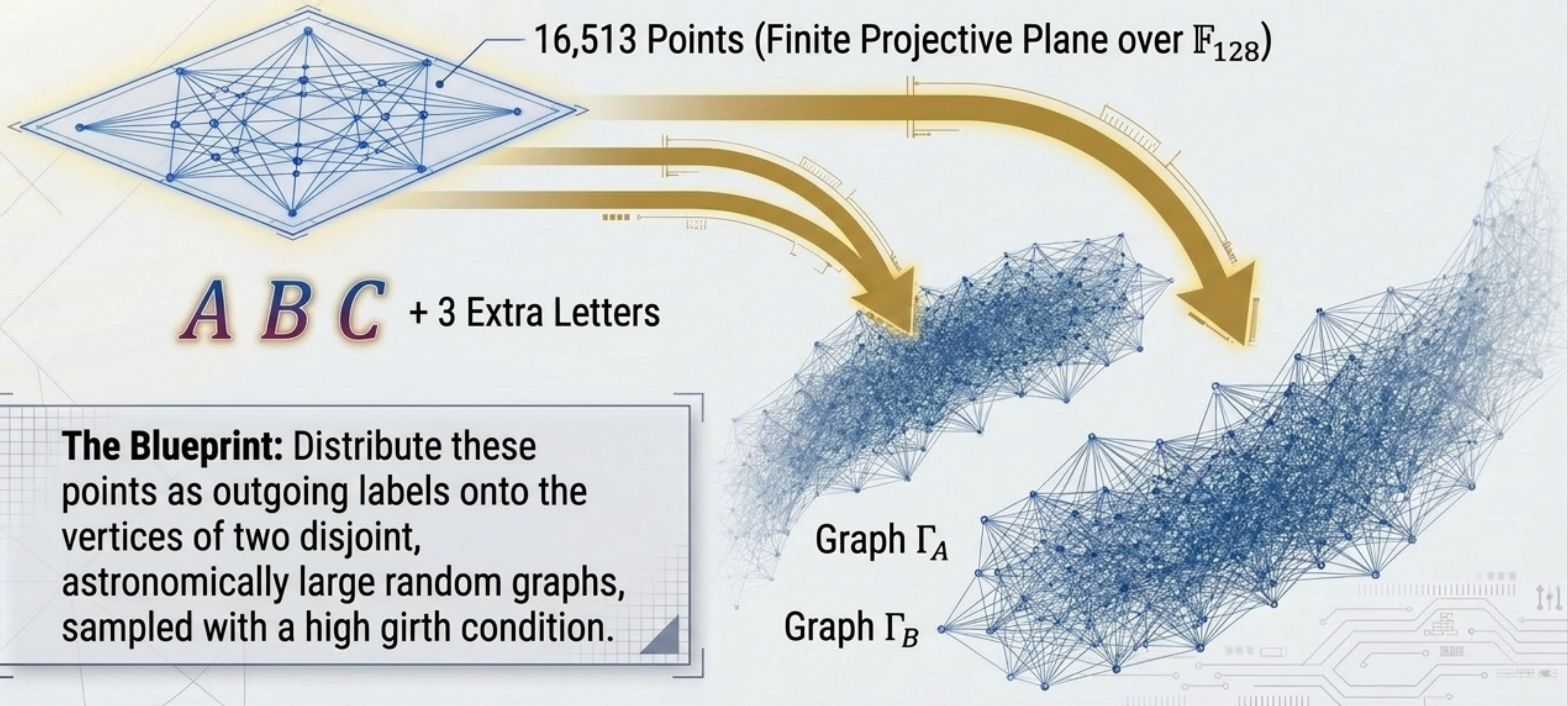
- Find a finitely presented torsion-free group G .
- Find non-zero elements $\alpha, \beta \in \mathbb{F}_2[G]$.
- Ensure their product perfectly self-annihilates.

The Geometric Mechanism



- Build a finite two-dimensional classifying space.
- Map out astronomical random graphs.
- Enforce strict topological rules to safely house the algebraic anomaly.

The Ingredients: Finite Projective Planes and Random Graphs



The Engine of Cancellation: Forced Odd Parity

The Geometric Axiom:
In a projective plane,
any two distinct lines
will always intersect at
exactly one point.

Exactly ONE Glowing
Point of Intersection

The Algebraic Application:
By assigning these line
classes as labels to graph
vertices, any cross-pair of
vertices (Side A to Side B)
is forced to share an odd
number of labels. Odd
parity is hard-coded into
the geometry, regardless of
how edges are matched.

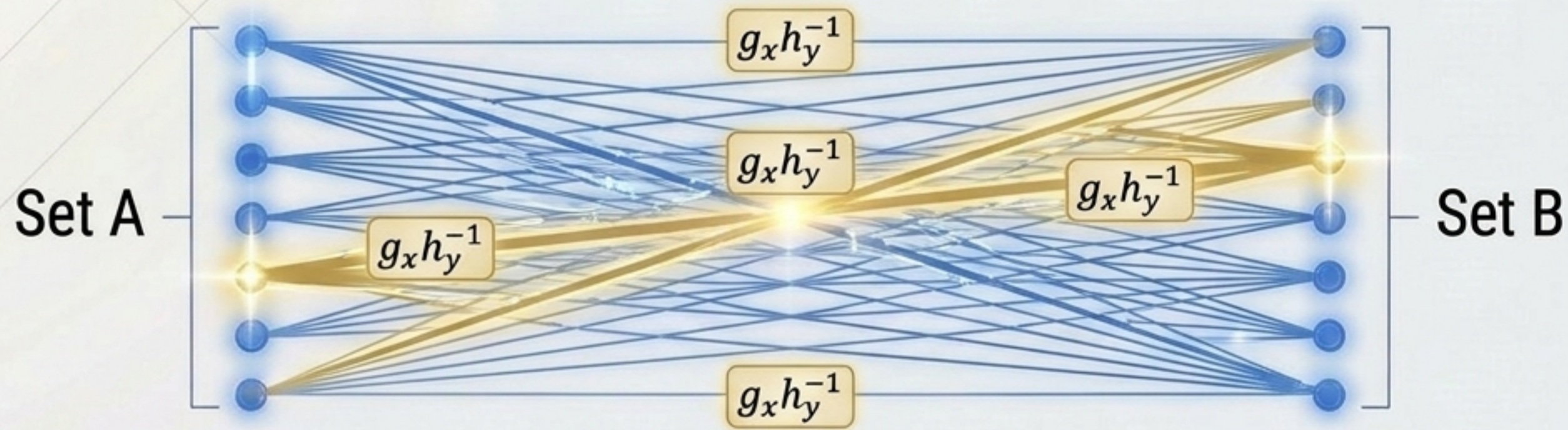
Node A

Node B

$$\langle \dots | \in [\beta]$$

$$j - \sum_{i=0}^n \{ \dots \} \quad \mathbb{I}+1$$

The Mod 2 Handshake: Perfect Algebraic Cancellation



1. Odd Degrees

Every vertex in the cross-paired graph has an odd degree (forced by the projective plane).

2. Even Components

By the Handshake Lemma (degree-sum formula), a graph where every vertex has an odd degree must have components with an EVEN number of vertices.

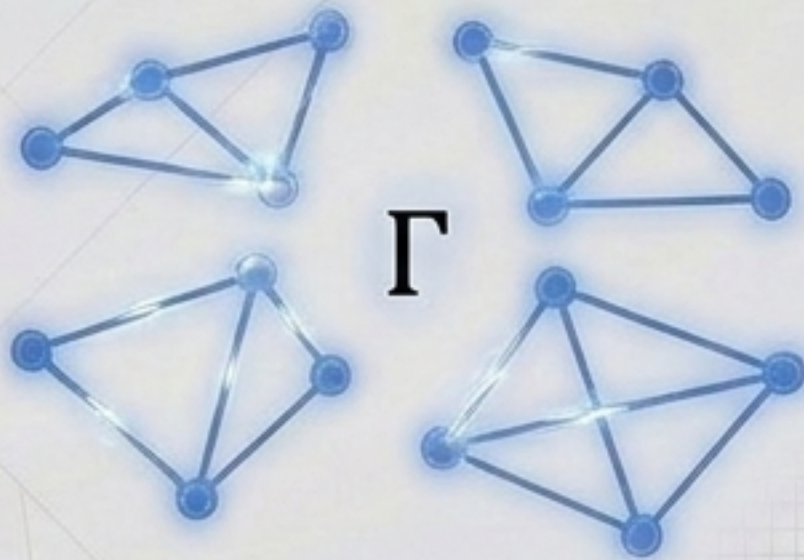
3. Annihilation

In characteristic 2 ($\mathbb{F}_2$), an even number of identical terms cancel out to 0.

Result: $\alpha\beta = 0$

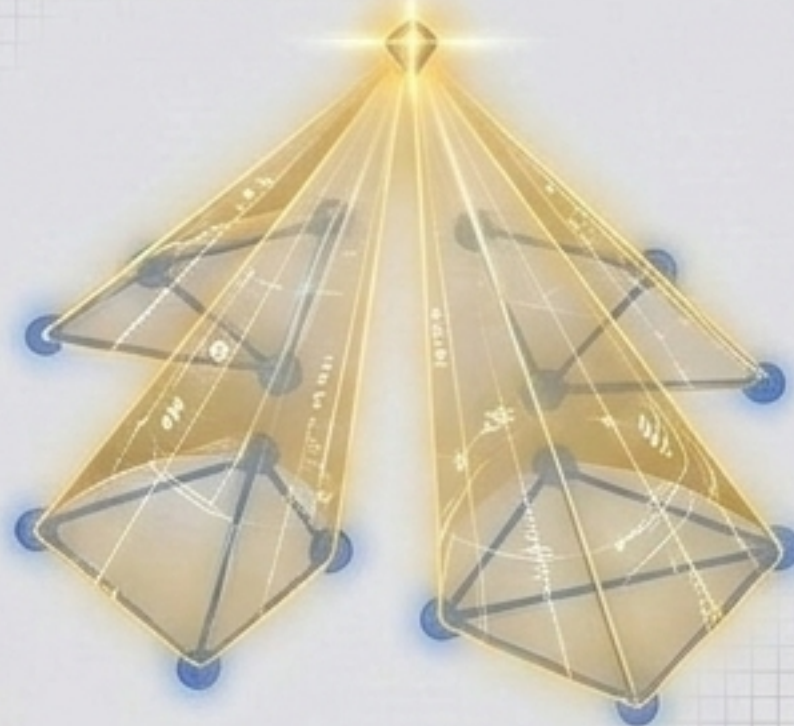
Topological Transformation: Forging a Group from Graphs

Start with Graphs



Take the disjoint graph components (Γ).

Attach Cones



Attach an abstract cone to each component. Coning geometrically “kills” the labels of all closed graph paths.

The 2-Complex



Result: A 2-complex X . The fundamental group $G = \pi_1(X)$. Path labels are now well-defined, independent of the route.

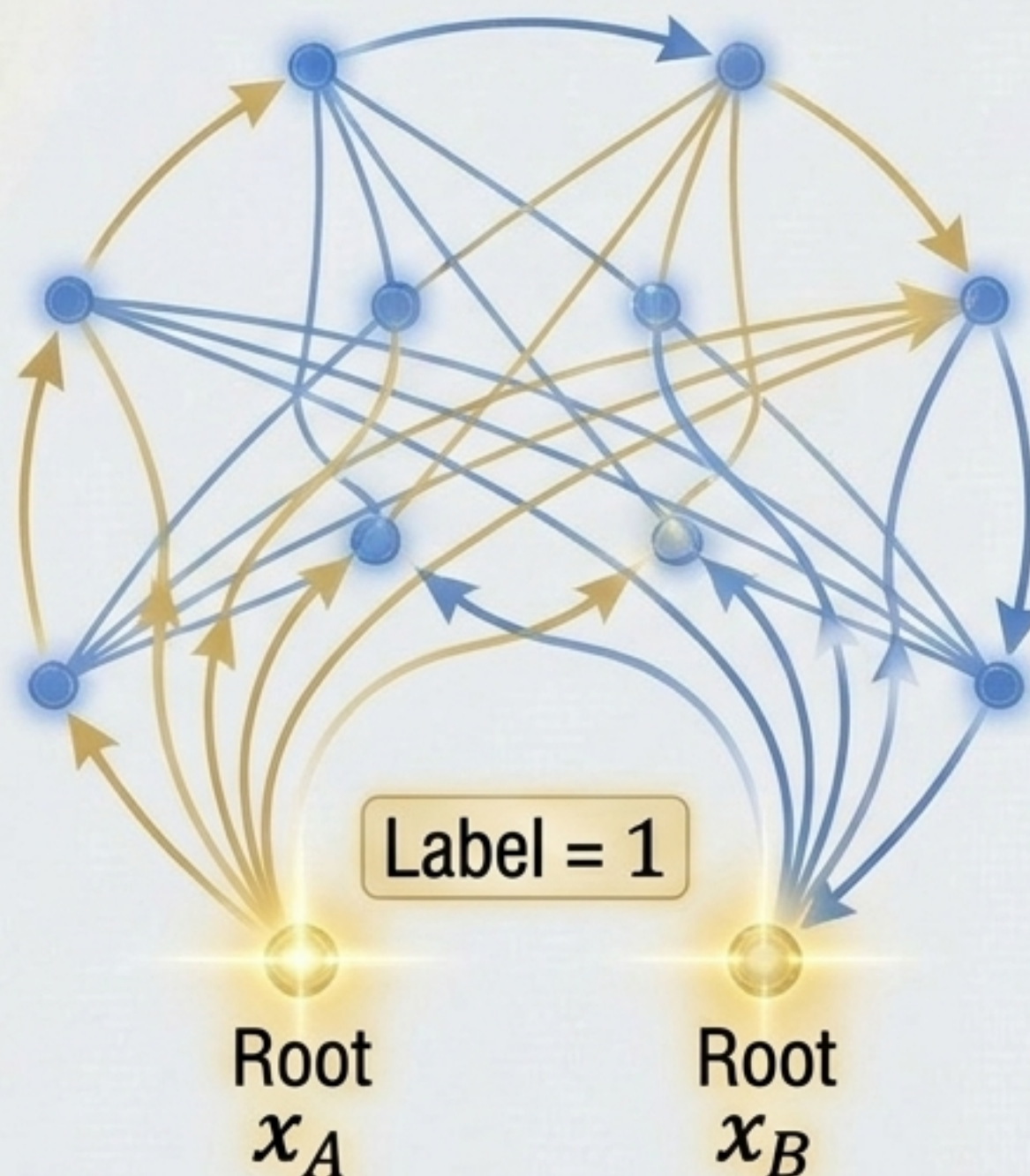
Safeguard I: Guaranteeing Non-Zero Factors

The Danger:

If everything cancels so perfectly, how do we know α and β weren't just equal to 0 from the start?

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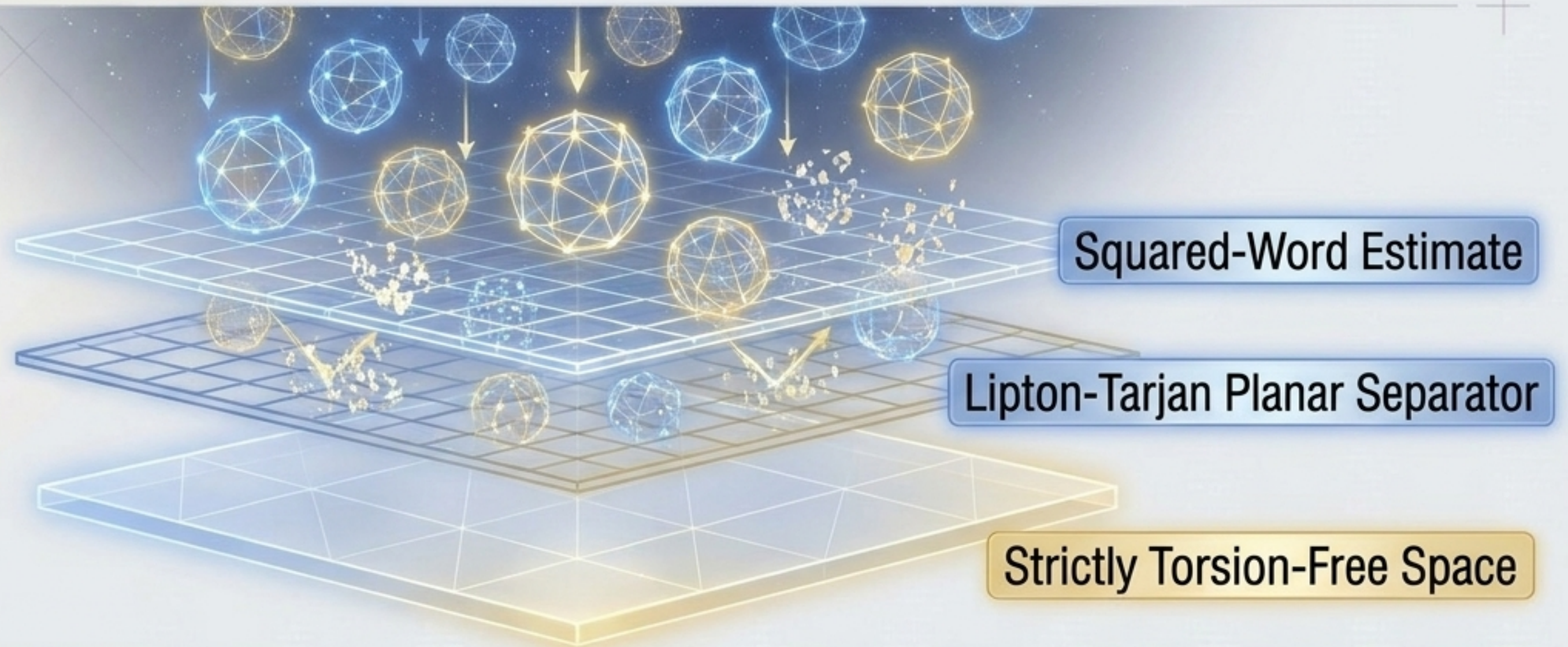
Root Isolation:

Only the designated roots of the graph carry the label 1.

Topological path routing guarantees that paths from a root to any other vertex have a nontrivial label. Therefore, only the root contributes to the identity coefficient.

Conclusion: α and β are strictly non-zero.

Safeguard II: The Torsion-Free Shield



The Final Boss: The group must be torsion-free. By excluding bounded systems of almost completely paired paths, the proof guarantees no reduced spherical arrangements can exist.

No spheres = Aspherical space ($\pi_2(X) = 0$) = Strictly Torsion-Free.

The Architecture of the Impossible

Characteristic 2 Algebra ($\mathbb{F}_2$)

Parity math allowing even identical terms to self-annihilate.

Algebraic Topology

2-Complex Asphericity and cones guaranteeing the space remains strictly torsion-free.

Theorem 1.1 Verified:
A Torsion-Free Group
with Zero Divisors.

Finite Geometry

$\mathbb{F}_{128}$ Projective Planes
guaranteeing odd-degree
intersections to trigger
the Handshake
Lemma.

The New Frontier: Formalized Verification at Scale

Higman's Conjecture: $G = F + T$

$$G = F + T$$

Kaplansky's Zero Divisor Conjecture:

G has no torsion.



$\mathbb{Z}[G]$ zero divisors?

Lets, the group leave in $\mathbb{R}^+$

How holes is G groups if $\mathbb{Z}[G]$

```
theorem torsion_free_zero_divisors :  
  ∀ (G : Type*) [Group G] [TorsionFree G],  
  ZeroDivisors (ℤ[G]) → False  
:= by  
  ...  
  for i in bastion G : TorsionFree  
  teaC : [Group G]  
  : ZeroDivisors [ℤ] = False  
  
dtheorem tenable_into(ℤ[G], ε)  
  if (α, zero => False) := by  
  := False  
  ...
```

This breakthrough marks more than the fall of an 80-year-old conjecture. It represents the rise of machine-checked mathematics. By utilizing astronomically large, probabilistically generated structures verified entirely in Lean, AI and formalization are scaling human mathematical intuition into entirely new dimensions.